Project Proposal – WSSOA (IT644)

Autumn 2026

Project Title :

PriceWise - Compare, Analyze, Shop Smarter

A web-based AI-Powered Smart Shopping Assistant

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Submitted By:

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Submitted To:

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Project Title and Short Description

Title :

PriceWise - A Shopping Intelligence Platform

Description :

PriceWise Shopping Intelligence Platform is an AI-powered platform designed to improve the online shopping experience by combining product discovery, comparison, price intelligence, and personalized shopping insights in a single platform.

With the rapid growth of e-commerce, users often find it difficult to choose the right product from thousands of available options across multiple online stores. They need to manually visit different platforms to compare specifications, check prices, analyze reviews, and track price changes. This process is time-consuming and does not always provide enough information to make an informed choice.

PriceWise solves this problem by providing a unified platform where users can search products, explore detailed specifications, compare similar products, and view prices from multiple stores such as Amazon, Flipkart, Croma, and other available sources. The platform also maintains historical price data and provides interactive price trend analysis to help users understand product pricing patterns.

Using Artificial Intelligence and Machine Learning, PriceWise provides advanced features such as future price prediction, personalized product recommendations, smart alternative suggestions, and price-drop alerts. These features help users discover suitable products, identify better deals, and gain meaningful insights before purchasing.

The goal of PriceWise is to transform traditional online shopping into a smarter, data-driven experience by combining product comparison, market analysis, and AI-powered recommendations on a single platform.

Application Type

Application Category: Web Application:

PriceWise is a web-based application that provides users with a centralized platform for product discovery, comparison, price tracking, and AI-powered shopping insights.

Reasons for Choosing Web Application:

- **Easy Accessibility:** Users can access PriceWise through any web browser without installing additional software.
- **Better Data Visualization:** The platform involves displaying product comparisons, price trends, historical graphs, and analytical insights, which can be effectively represented through a web interface.
- **Cross-Platform Support:** Users can access the application from different devices such as laptops, tablets, and smartphones.
- **Easy Maintenance and Updates:** New features, improvements, and updates can be deployed easily without requiring users to manually update the application.
- **Scalability:** The web architecture allows smooth integration with backend services, databases, and AI/ML models as the platform expands.
- **Integration Capability:** The application can easily connect with external APIs, data sources, notification systems, and machine learning services.

Motivation / Problem Statement

Problem Statement:

The rapid growth of e-commerce has made online shopping more convenient but has also increased the complexity of choosing the right product. Users have access to thousands of products across multiple platforms, making it difficult to compare options and identify the best choice.

Currently, users need to manually visit different websites to:

- Search for products across multiple platforms.
- Compare product specifications and features.
- Check prices from different online stores.
- Analyze whether the current price is reasonable.
- Track historical price changes.
- Find better alternatives within their budget.

This process is time-consuming and often requires users to collect information from multiple sources before making a purchase decision. Existing shopping platforms mainly focus on selling products and provide limited insights about price trends, future price changes, and personalized recommendations.

Motivation:

The motivation behind PriceWise is to create a unified platform that simplifies the online shopping process by combining product information, price comparison, historical analysis, and AI-powered insights in one place.

The platform aims to:

- Reduce the effort required to research products across multiple websites.
- Help users compare products based on features, specifications, and prices.
- Provide insights into price trends and possible future price movements.
- Recommend suitable products based on user preferences and requirements.
- Help users discover better alternatives and available deals.

By combining data analytics and machine learning, PriceWise aims to make online shopping more intelligent, transparent, and efficient.

Target Users and Use Cases

Use Cases :

Use Case	Description
Product Search	Search products using name, category, and brand filters.
Product Comparison	Compare specifications, features, ratings, and prices of multiple products.
Price Comparison	Compare prices across different online stores to find better deals.
Price Analysis	View historical price trends and analyze price changes.
AI Insights	Get price predictions, recommendations, and smart alternatives.
Price Alerts	Receive notifications when product prices drop or reach the desired value.

Target Users:

Target Users	Use Cases
Online Shoppers	Search products, compare options, check prices, and find better deals before purchasing.
Budget-Conscious Users	Discover products within budget and find cost-effective alternatives.
Students & Professionals	Find suitable products based on requirements such as performance, features, and price.
Technology Enthusiasts	Track new products, analyze price trends, and monitor market changes.

Scope and High-Level Features

Scope of the Project:

The project aims to develop a unified shopping platform that helps users discover products, compare options, analyze prices, and receive intelligent shopping insights.

The system reduces the need to visit multiple shopping websites by bringing product details, price comparison, historical trends, and recommendations together in one place. The scope includes product search, price tracking, AI-based analysis, and personalized shopping features.

High-Level Features:

1. Product Discovery

- Search products by name, category, and brand.
- View specifications, images, and ratings.

2. Product Comparison

- Compare products based on features, specifications, and prices.
- Highlight key differences between products.

3. Price Intelligence

- Compare prices across multiple stores.
- Track historical price trends using graphs.

4. AI-Powered Insights

- Predict future price trends.
- Provide smart recommendations and alternative product suggestions.

5. User Personalization

- Wishlist and watchlist management.
- Personalized recommendations based on user preferences.

6. Price Alerts

- Notify users about price drops and better deals.

7. Data Analytics

- Analyze product and pricing data to generate useful shopping insights.

Technology Stack

Category	Technologies
Frontend	React.js, Tailwind CSS, Shadcn/UI, Recharts / Chart.js
Backend	Node.js / Python
Database	PostgreSQL
Cache	Redis
AI/ML	Python, Pandas, NumPy, Scikit-learn, Prophet / XGBoost
ML API Service	FastAPI
Authentication	JWT, Google OAuth (Optional)
API Communication	REST APIs, Axios
Deployment	Vercel (Frontend), Render / Railway / AWS (Backend & ML Service), Neon PostgreSQL (Database)
Version Control	Git, GitHub

Roles and Responsibilities

Responsibility	Team Members
Frontend Development & UI Implementation	Divyesh Dandwani, Harshil Dodwani
Backend Development & API Integration	Diya Jain, Aman Rohera, Divyesh Dandwani
AI/ML Development & Recommendation System	Piyush Tanwani, Priya Shah, Yash Gangwani, Harshil Dodwani
Authentication & User Management	Yash Gangwani, Diya Jain
Database Design & Management	Piyush Tanwani, Vivek Dhanwani
Data Collection & External API Integration	Vivek Dhanwani, Aman Rohera
Testing, Documentation & Deployment	Diya Jain, Harshil Dodwani, Priya Shah

Timeline with Milestones

Week	Milestone	Activities
Week 1	Project Proposal	Finalize project scope, team roles, and submit the proposal.
Week 2	Planning & Requirements	Prepare project plan, gather requirements, create user stories and initial wireframes.
Week 3	System Design	Design database, system architecture, APIs, and UI mockups.
Week 4–7	Core Development	Develop frontend, backend APIs, authentication, and product management modules.
Week 8–10	Feature Integration	Implement product comparison, price tracking, historical graphs, and AI recommendation features.
Week 11	Mid-Sem Demo	Demonstrate a working prototype with core functionalities.
Week 12–14	Testing & Refinement	Perform testing, fix bugs, optimize performance, and complete documentation.
Week 15	Deployment	Deploy the application and integrate all modules.
Week 16	Final Submission	Submit the final project report and present the completed application.